Joined_Data

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#Packages

```r
# joined csv files from shared drive with FIPs
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.1     v readr     2.1.5
## v ggplot2   3.5.2     v stringr   1.6.0
## v lubridate 1.9.4     v tibble    3.2.1
## v purrr     1.0.4     v tidyr     1.3.1

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become error
```

```r
library(readr)
library(readxl)
```

#Census Tract Data and Unique IDS

City Data and Tract Overlap Data

#Joining Data

#Loading the new Dataset

```r
#this is me getting the frequency of City Names
table(LACityData$City_Name)
```

```
##
##          Agoura Hills              Alhambra               Arcadia
##                  1290                 11121                  5612
##               Artesia                Avalon                 Azusa
##                  1013                   363                  5759
##          Baldwin Park                  Bell          Bell Gardens
##                  7021                  3939                  3913
##            Bellflower         Beverly Hills               Burbank
##                  7743                  3017                 10088
##             Calabasas                Carson              Cerritos
##                  2006                  5126                  2290
##             Claremont              Commerce               Compton
##                  2387                  1222                  9763
##                Covina                Cudahy           Culver City
##                  4698                  2744                  3856
##           Diamond Bar                Downey                Duarte
##                  3081                  8707                  1838
##              El Monte            El Segundo               Gardena
##                 12307                  1926                  7956
##              Glendale              Glendora      Hawaiian Gardens
##                 20269                  3946                  1142
##             Hawthorne        Hermosa Beach       Huntington Park
##                 10196                  2177                  6915
##             Inglewood  La Canada Flintridge      La Habra Heights
##                 13270                   665                   379
##             La Mirada             La Puente              La Verne
##                  2277                  3048                  1898
##              Lakewood             Lancaster              Lawndale
##                  4631                  8598                  3679
##                Lomita            Long Beach           Los Angeles
##                  1996                 45020                446899
##               Lynwood                Malibu       Manhattan Beach
##                  5029                  1570                  2862
##               Maywood              Monrovia            Montebello
##                  3187                  4789                  7101
##         Monterey Park               Norwalk              Palmdale
##                  6707                  5403                  6848
##  Palos Verdes Estates             Paramount              Pasadena
##                   480                  5450                 16085
##           Pico Rivera                Pomona   Rancho Palos Verdes
##                  3922                 12360                  1915
##         Redondo Beach Rolling Hills Estates              Rosemead
##                  6457                   989                  6606
##             San Dimas          San Fernando           San Gabriel
##                  2561                  1904                  5049
##            San Marino         Santa Clarita      Santa Fe Springs
##                   599                 13010                  2422
##          Santa Monica          Sierra Madre           Signal Hill
##                 10047                   611                  1233
##         South El Monte            South Gate        South Pasadena
```

```
##                2545                9570                2744
##         Temple City            Torrance              Walnut
##                4570               10562                1479
##         West Covina     West Hollywood    Westlake Village
##                8019                4202                 894
##            Whittier
##                8539
```

Random Forest

```r
library(randomForest)
```

```
## randomForest 4.7-1.2
```

```
## Type rfNews() to see new features/changes/bug fixes.
```

```
##
## Attaching package: 'randomForest'
```

```
## The following object is masked from 'package:ggplot2':
##
##     margin
```

```
## The following object is masked from 'package:dplyr':
##
##     combine
```

```r
set.seed(123)
energy_rf <- energy_data %>%
  select(energy_burden_cat, trees, lakeTF, parks_in_city, avg_acres_in_city) %>% na.omit() %>%
  group_by(energy_burden_cat) %>% # stratified sampling accounts for uneven distribution of cats
  sample_n(20000) %>%
  ungroup()
energy_model <- randomForest(
  energy_burden_cat ~ trees + lakeTF + parks_in_city + avg_acres_in_city,
                           data = energy_rf,
                           importance = TRUE,
                           ntree = 300)
print(energy_model)
```

```
##
## Call:
##  randomForest(formula = energy_burden_cat ~ trees + lakeTF + parks_in_city +      avg_acres_in_city,
##                Type of random forest: classification
##                      Number of trees: 300
## No. of variables tried at each split: 2
##
##         OOB estimate of  error rate: 65.17%
## Confusion matrix:
```

```
##          Low High Severe class.error
## Low    17789  814   1397     0.11055
## High   17180 1267   1553     0.93665
## Severe 16956 1202   1842     0.90790
```

```r
importance(energy_model)
```

```
##                         Low     High     Severe MeanDecreaseAccuracy
## trees              6.332807 5.342006  -7.305821            20.517239
## lakeTF             4.153771 2.936927  -7.214415             4.611315
## parks_in_city      2.926863 3.690881  -3.606382            16.670914
## avg_acres_in_city 12.550726 4.064728 -11.510652            30.860363
##                   MeanDecreaseGini
## trees                    47.291124
## lakeTF                    5.454949
## parks_in_city            38.132667
## avg_acres_in_city        92.712657
```

Severe energy burden is predicted the best out of the three categories (based on lowest class error) HINCP.UNITS (average annual household income) is the strongest predictor of energy burden classification (91.96). Water body presence is the least important predictor (13.17). Energy burden is heavily associated with economic factors, while environmental features contributing less.

Descriptives of Variables

```r
library(skimr)
```

```
## Warning: package 'skimr' was built under R version 4.5.2
```

```r
library(dplyr)

energy_data %>%
  select(energy_burden_cat, trees, lakeTF, parks_in_city, avg_acres_in_city) %>%
  skim()
```

Table 1: Data summary

Name	Piped data
Number of rows	719384
Number of columns	5
Column type frequency:	
factor	1
numeric	4
Group variables	None

Variable type: factor

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
energy_burden_cat	0	1	FALSE	3	Low: 638240, Hig: 42053, Sev: 39091

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
trees	237123	0.67	138.91	460.74	1.0	2.00	2.00	2.00	3858.00	
lakeTF	0	1.00	0.75	0.44	0.0	0.00	1.00	1.00	1.00	
parks_in_city	4585	0.99	240.79	199.30	2.0	21.00	433.00	433.00	433.00	
avg_acres_in_city	4585	0.99	36.77	106.83	0.5	14.16	29.79	29.79	6023.29	

```r
energy_data %>%
  group_by(City_Name) %>%
  summarise(
    avg_trees = mean(trees, na.rm = TRUE),
    sd_trees = sd(trees, na.rm = TRUE),
    min_trees = min(trees, na.rm = TRUE),
    max_trees = max(trees, na.rm = TRUE),

    avg_water = mean(lakeTF, na.rm = TRUE),
    sd_water = sd(lakeTF, na.rm = TRUE),
    min_water = min(lakeTF, na.rm = TRUE),
    max_water = max(lakeTF, na.rm = TRUE),

    avg_parks = mean(parks_in_city, na.rm = TRUE),
    sd_parks = sd(parks_in_city, na.rm = TRUE),
    min_parks = min(parks_in_city, na.rm = TRUE),
    max_parks = max(parks_in_city, na.rm = TRUE),

    avg_acres = mean(avg_acres_in_city, na.rm = TRUE),
    sd_acres = sd(avg_acres_in_city, na.rm = TRUE),
    min_acres = min(avg_acres_in_city, na.rm = TRUE),
    max_acres = max(avg_acres_in_city, na.rm = TRUE)

  )
```

```
## Warning: There were 140 warnings in `summarise()`.
## The first warning was:
## i In argument: `min_trees = min(trees, na.rm = TRUE)`.
## i In group 1: `City_Name = "AGOURA HILLS"`.
## Caused by warning in `min()`:
## ! no non-missing arguments to min; returning Inf
## i Run `dplyr::last_dplyr_warnings()` to see the 139 remaining warnings.
```

```
## # A tibble: 82 x 17
##    City_Name avg_trees sd_trees min_trees max_trees avg_water sd_water min_water
##    <chr>         <dbl>    <dbl>     <dbl>     <dbl>     <dbl>    <dbl>     <dbl>
##  1 AGOURA H~       NaN       NA       Inf      -Inf         1        0         1
##  2 ALHAMBRA        NaN       NA       Inf      -Inf         0        0         0
```

```
##   3 ARCADIA        1508        0      1508      1508          1         0          1
##   4 ARTESIA         NaN       NA       Inf      -Inf          0         0          0
##   5 AVALON          NaN       NA       Inf      -Inf          0         0          0
##   6 AZUSA          2613        0      2613      2613          1         0          1
##   7 BALDWIN ~       NaN       NA       Inf      -Inf          0         0          0
##   8 BELL            NaN       NA       Inf      -Inf          0         0          0
##   9 BELL GAR~       NaN       NA       Inf      -Inf          1         0          1
##  10 BELLFLOW~       NaN       NA       Inf      -Inf          0         0          0
## # i 72 more rows
## # i 9 more variables: max_water <dbl>, avg_parks <dbl>, sd_parks <dbl>,
## #   min_parks <dbl>, max_parks <dbl>, avg_acres <dbl>, sd_acres <dbl>,
## #   min_acres <dbl>, max_acres <dbl>
```

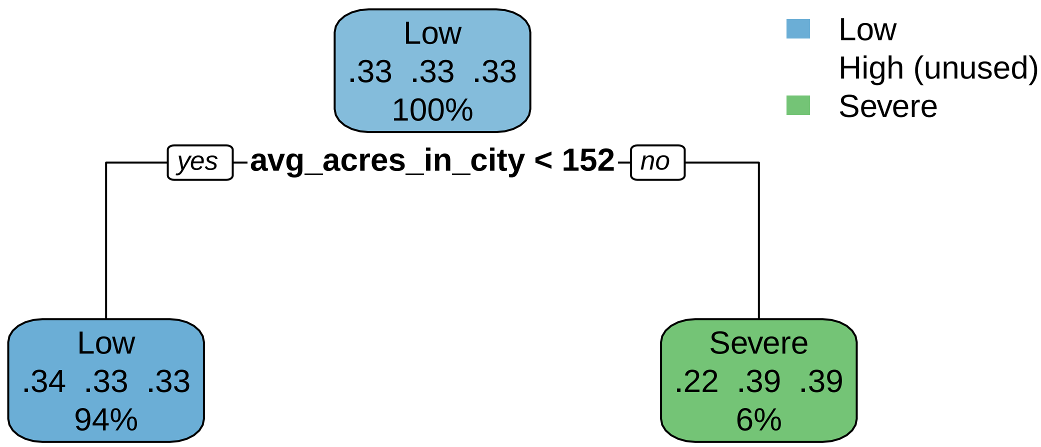
Node Visual

```r
library(rpart)
library(rpart.plot)
```

```
## Warning: package 'rpart.plot' was built under R version 4.5.2
```

```r
set.seed(123)

tree_visual <- rpart(
  energy_burden_cat ~ trees + lakeTF + parks_in_city + avg_acres_in_city,
  data = energy_rf,
  method = "class"
)
rpart.plot(tree_visual)
```


Low
.33 .33 .33
100%
yes
avg_acres_in_city < 152
no
Low
.34 .33 .33
94%
Severe
.22 .39 .39
6%
Low
High (unused)
Severe